

DIFFERENT CASES OF EXECUTING TASK :

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1. Perform a Single Task from a Single Thread

- One thread performs **only one task**.

```
package com.string;

class MyThread extends Thread {
    public void run() {
        System.out.println("Single task from a single thread.");
    }
}

public class Main {
    public static void main(String[] args) {
        MyThread t1 = new MyThread();
        t1.start(); // Runs a single task in a single thread
    }
}
```

2. Perform a Single Task from Multiple Threads

- Multiple threads execute the **same task**.

```
package com.string;

class MyThread extends Thread {
    public void run() {
        System.out.println("Single task executed by: " + Thread.currentThread().getName());
    }
}

public class Main {
    public static void main(String[] args) {
        MyThread t1 = new MyThread();
        MyThread t2 = new MyThread();
        MyThread t3 = new MyThread();

        t1.start();
        t2.start();
        t3.start();
    }
}
```

- How many thread creates this time ?
- 4 thread created .. (main is also a thread created by JVM).

3. Perform Multiple Tasks from a Single Thread

- A **single thread** performs multiple tasks sequentially.
- This is not possible
- Ex : VLC perform tasks like playing video , song , progress bar so in this case single thread performs first playing the video part then after it play song so again it is not the correct way.

```
package com.string;

class Task {
    void task1() {
        System.out.println("Task 1 is running.");
    }
    void task2() {
        System.out.println("Task 2 is running.");
    }
}

public class Main {
    public static void main(String[] args) {
        Task obj = new Task();
        obj.task1(); // First task
        obj.task2(); // Second task
    }
}
```

4. Perform Multiple Tasks from Multiple Threads

- Multiple threads execute different tasks.

```
package com.string;

class Task1 extends Thread {
    public void run() {
        System.out.println("Task 1 executed by: " + Thread.currentThread().getName());
    }
}

class Task2 extends Thread {
    public void run() {
        System.out.println("Task 2 executed by: " + Thread.currentThread().getName());
    }
}

public class Main {
    public static void main(String[] args) {
        Task1 t1 = new Task1();
        Task2 t2 = new Task2();

        t1.start();
        t2.start();
    }
}
```